Abhishek Surendra Shamkuwar

M.Tech., Signal Processing and Machine Learning | Dr. B. R. Ambedkar NIT Jalandhar, India
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RESEARCH INTERESTS

I'm interested in using machine learning, computer vision, and signal processing to build AI that is actually useful to people. Healthcare is where most of my work has been so far. For my thesis, I built a model that predicts sepsis early from ICU data, which meant combining different kinds of patient data, like vital signs and lab results, in a single model. Next, I want to work with brain signals (EEG) and medical images such as CT and MRI. I'm also drawn to human–computer interaction and robotics, and I'm curious how the same ideas could be used there, for example in brain-computer interfaces or robots that respond to people. Wherever I end up, I'd like to learn how to make these models accurate and something people can trust.

EDUCATION

M.Tech. in Signal Processing and Machine Learning, Dr. B. R. Ambedkar National Institute of Technology Jalandhar, Punjab, India — 2024 – 2026
CGPA: 8.38/10. Thesis: *Early Sepsis Prediction Using Multimodal Deep Learning*. Supervisor: Dr. Manjeet Singh.
Coursework: Machine Learning, Applied Machine Learning, Deep Learning, Statistical Signal Processing, Advanced Signal Processing, Probability and Stochastic Processes, Applied Linear Algebra and Optimization.

B.Tech. in Electronics and Communication Engineering, Indian Institute of Information Technology (IIIT) Sri City, Chittoor, Andhra Pradesh, India — 2020 – 2024
Coursework: Pattern Recognition, Digital Signal Processing, Digital Image Processing, Data Structures and Algorithms, Object-Oriented Programming, Embedded Systems, Control Systems.

PUBLICATIONS

Journal

[J1] A. Kar, P. K. Shill, **A. S. Shamkuwar**, S. Saini, Vasundhara, S. Shoba, and M. G. Christensen, "Adaptive Improved Primal-Dual Distributed Stochastic Gradient Driven Distributed Active Noise Control System," *IEEE Open Journal of Signal Processing*, vol. 7, pp. 643–658, 2026. doi: 10.1109/OJSP.2026.3701314

My role: wrote the manuscript up to the simulation section (introduction, background, proposed algorithm and performance analysis) from the methodology provided by my supervisor, and made Figures 1 and 2.

Conference

[C1] **A. S. Shamkuwar** (first author), P. Bhardwaj, N. Kumar, and M. Singh, "Early Sepsis Detection from ICU Data Using Multimodal Deep Learning," in *2026 Int. Conf. on Intelligent Processing, Hardware, Electronics, and Radio Systems (CIPHER)*, Jalandhar, India, Feb. 2026, pp. 1–5. doi: 10.1109/CIPHER70417.2026.11523895

Presented at the conference.

[C2] A. Sharma, **A. S. Shamkuwar**, U. Anand, R. Kumar, N. Kumar, and M. Singh, "Spatio-Temporal Node Localization in Wireless Sensor Networks Using Hybrid Recurrent-Graph Neural Network," in *2026 CIPHER*, Jalandhar, India, Feb. 2026, pp. 1–5. doi: 10.1109/CIPHER70417.2026.11524038

My role: ran the simulations and produced the results (four graph and recurrent models across 15 mobility and noise settings).

RESEARCH EXPERIENCE

M.Tech. Thesis: MultimodalSepsisNet (early sepsis prediction, PhysioNet/CinC 2019) — Jun 2025 – May 2026
Supervisor: Dr. Manjeet Singh, NIT Jalandhar

- Designed a multimodal network that splits 40 clinical variables into four modalities (vital signs, laboratory values, missingness patterns, static demographics), with TCN–BiGRU, BiGRU and feedforward encoders fused by eight-head cross-modal attention (3.0M parameters).
- Built a leakage-free pipeline on 39,721 ICU records: patient-level stratified splits, imputation, normalisation and class-imbalance handling, evaluated at 6-hour and 12-hour prediction horizons.
- Achieved the highest AUC-ROC and AUPRC among eight baselines (LR, DT, RF, XGBoost, RNN, LSTM, GRU, TCN) at both horizons: 0.9535 and 0.7740 at 12 h (+0.087 AUPRC over XGBoost) and 0.9407 and 0.7441 at 6 h, with 95% bootstrap confidence intervals.
- Ablations showed that removing the laboratory stream, missingness stream or cross-modal attention lowers AUPRC by 0.224, 0.083 and 0.069; attention analysis showed focus on early ICU hours and admission laboratory results.

PROJECTS

Human Activity Recognition: Signal Processing vs. Deep Learning (code) Python, PyTorch

- Engineered 63 time- and frequency-domain features (Welch PSD, zero-crossing rate, spectral entropy) from raw 9-channel accelerometer and gyroscope signals; compared Random Forest, 1D-CNN and bidirectional LSTM.
- Fixed a sitting/standing failure using feature-importance analysis and gravity-orientation features, raising accuracy from 89% to 95% (macro F1 0.94).
- Validated with leave-one-subject-out cross-validation over 30 subjects (93.9% ± 5.3% accuracy); deployed with FastAPI and Streamlit.

Real-Time Pothole Detection and Geotagging (B.Tech. final-year project; code) Python, OpenCV

- Built a real-time detector running a YOLO-tiny model through OpenCV DNN with CUDA FP16 inference, using confidence thresholding, non-maximum suppression and box-size filtering to suppress false detections.
- Saved annotated frames with location coordinates at fixed intervals and recorded result video with a live FPS overlay.

Finance RAG System over SEC EDGAR Filings Python, FastAPI, React

- Built retrieval-augmented question answering over 10-K/10-Q filings: automated ingestion and cleaning, tables kept as single chunks, hybrid BGE/FAISS and BM25 retrieval with reciprocal rank fusion, and citation-grounded Gemini answers.
- Served through a Dockerized FastAPI API and React interface; wrote routing and retrieval evaluation scripts.

TECHNICAL SKILLS

Programming: Python, MATLAB, C/C++

Machine learning: PyTorch, PyTorch Geometric, scikit-learn, XGBoost; CNN/TCN, RNN/LSTM/GRU, graph neural networks (GCN, GAT), attention-based multimodal fusion

Computer vision: OpenCV, YOLO-based object detection

Signal processing and health data: Spectral analysis (Welch PSD), time- and frequency-domain features for accelerometer and gyroscope signals; clinical (ICU) time-series with missing data

Data and evaluation: Pandas, NumPy, Matplotlib; AUROC/AUPRC, bootstrap confidence intervals, cross-validation, class-imbalance handling

Tools: Git, Docker, FastAPI

REFERENCES

Available on request.